

Chapter 5

Reaction Kinetics

Learning Objectives

- Explain reaction rate and factors affecting reaction rate
- Explain the rate law and the order of reactions (zero-, first-, second-order)
- Integrated rate law and reaction order
- Describe reaction half-life
- Draw energy profile diagrams and indicate clearly the role of catalyst on the activation energy
- Calculate activation energy and rate constant by Arrhenius equation

Part 1

Reaction Rate and factors affecting rate

The wide range of reaction rates



slow



fast



very fast

Chemical Kinetics

- Definition: the study of reaction rates, or the changes in the concentrations of reactants and products with time
- By studying kinetics, insights are gained to manipulate reaction conditions to achieve desired outcomes
- One of the main goals of chemical kinetics is to understand reaction mechanisms. This allows us to find ways to control reactions

Controlling and Using Chemical Reactions

Seven questions:

Thermodynamics

- How much energy is absorbed/evolved by the reaction?
- Does the reaction proceed spontaneously, in a certain direction?
- How much useful work can be obtained from the reaction?
- What is the value of the equilibrium constant for the reaction?

Kinetics

- What is the rate of reaction?
- What factors **affect** the rate, and in what ways?
- What is the mechanism behind the reaction, and hence, what process limits the rate?

Reaction Rate: definition

- change in concentration of reactants or products per unit time
 - Reactant concentrations decrease while product concentrations increase. For example:



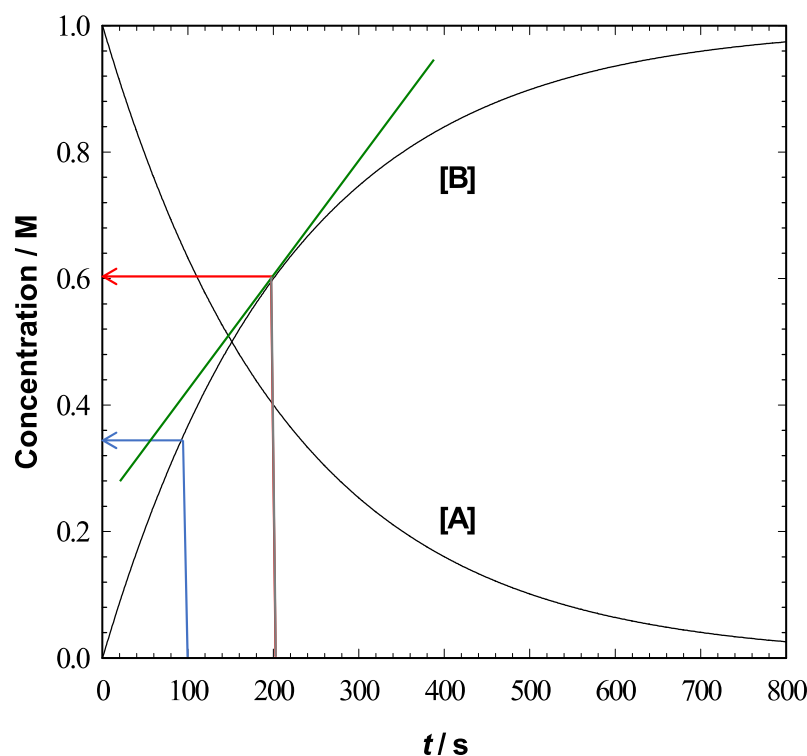
$$\text{rate of reaction} = - \frac{\text{change in concentration of A}}{\text{change in time}} = - \frac{\text{conc } A_2 - \text{conc } A_1}{t_2 - t_1}$$

(for reactant)

$$- \frac{\Delta(\text{conc } A)}{\Delta t} \quad \text{or} \quad - \frac{\Delta [A]}{\Delta t}$$

Note: here the change in concentration of A is negative, thus leading to a negative reaction rate. By convention, rate is defined as a positive quantity, while the negative or positive sign in the rate expression denotes a decrease or an increase in the reaction rate respectively.

Average Rate and Instantaneous Rate



Reaction: $A \rightarrow B$

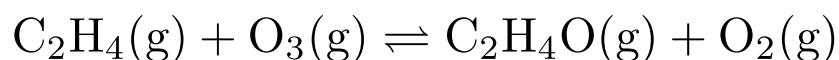
The average reaction rate for the **product** between the reaction times 100s and 200s can be calculated from the graph.

$$\begin{aligned} \text{Average Rate} &= \frac{\Delta[B]}{\Delta t} = \frac{[B]_{200\text{s}} - [B]_{100\text{s}}}{(200\text{s} - 100\text{s})} \\ \frac{\Delta[B]}{\Delta t} &= \frac{(0.61 - 0.37)}{(200 - 100)} \\ &= 2.4 \times 10^{-3} \text{Ms}^{-1} \end{aligned}$$

Initial rate: instantaneous rate when $t = 0$

The instantaneous rate at 200 s is from the slope of a tangent to the curve at that time. Why?
Is the instantaneous rate faster or slower than the average rate?

Another Example: Reaction of ethene with ozone



Time (s)	Concentration of O_3 (mol/L)	
0.0	3.20×10^{-5}	
10.0	2.42×10^{-5}	rate = $-\frac{\Delta [\text{C}_2\text{H}_4]}{\Delta t}$
20.0	1.95×10^{-5}	
30.0	1.63×10^{-5}	$-\frac{\Delta [\text{O}_3]}{\Delta t}$
40.0	1.40×10^{-5}	
50.0	1.23×10^{-5}	
60.0	1.10×10^{-5}	

Table 16.1 Principles of General Chemistry, Silberberg, McGraw-Hill Education (Asia)

Calculate the average rate between:

0.0 s and 60.0 s

0.0 s and 30.0 s

30.0 s and 60.0 s

How to find instantaneous rate from this data?

What is the rate if the stoichiometry is not 1?

- In most reactions, the coefficients of the reaction components are not the same.

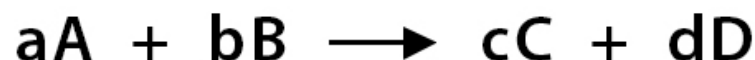


- For the reaction above, with every mole of H_2 used:
1 mole of I_2 will be consumed, to produce 2 moles of HI as product
- Therefore the rate of change of reactants and products will be different
(HI appears to be produced at double the rate)
- One reaction having two rates (0.03 for I_2/H_2 and 0.06 for HI) is confusing
- In order to be consistent, the changes in the concentration of each component is *multiplied by 1/coefficient*

$$\text{Rate} = -\frac{\Delta[\text{H}_2]}{\Delta t} = -\frac{\Delta[\text{I}_2]}{\Delta t} = +\left(\frac{1}{2}\right) \frac{\Delta[\text{HI}]}{\Delta t}$$

General formula for reaction rate

- In general, for the reaction



$$r = -\frac{1}{a} \frac{d[A]}{dt} = -\frac{1}{b} \frac{d[B]}{dt} = \frac{1}{c} \frac{d[C]}{dt} = \frac{1}{d} \frac{d[D]}{dt}$$

- Remember to divide by stoichiometric number
- Remember the sign: positive for product; negative for reactant
- In any case, the rate must be a positive value
- $\Delta[X]/\Delta t$ gives average rate; $d[X]/dt$ gives instantaneous rate

What affects reaction rate?

- Four factors that influence the reaction rate
 1. **Physical state** of the reactants – molecules must mix to collide
 2. **Concentration** – molecules must collide to react (pressure affects concentration of gases)
 3. **Temperature** – molecules must collide with enough energy to react
 4. Use of a **catalyst**

Physical state of reactants

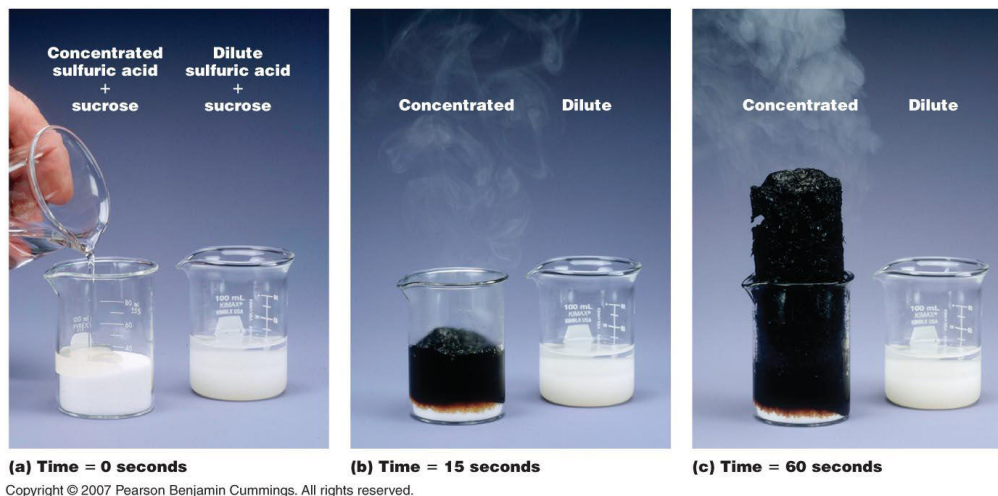
- Two substances cannot react with each other unless their constituent particles come into contact; if there is no contact, the rate of reaction will be zero.
- If reactants are in a *homogeneous* solution, random thermal motion brings them into contact.
- If the reactants are in two different phases (ie. *heterogeneous* phases), collisions between the reactants can occur only at the interface .
- Therefore, the number of collisions between reactants per unit time is limited, so is the reaction rate.
- The rate of heterogeneous reactions depend on the surface area of the more condensed phase.



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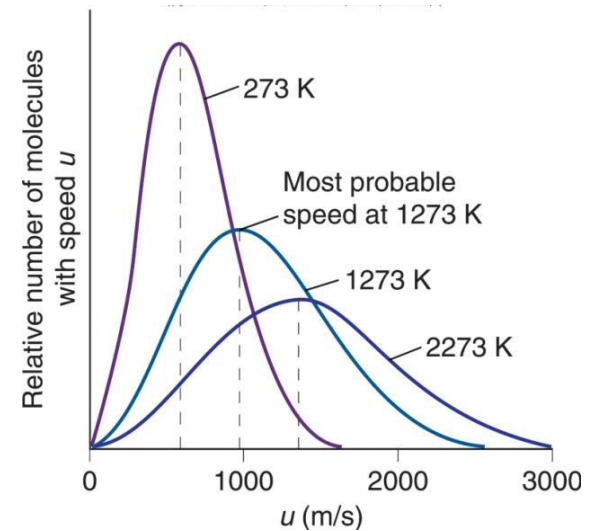
Concentration effects

- The more reactant particles are present in the container, the higher the probability of collision, the higher the probability of a reaction happening.
- Hence, the rate of reaction usually increases as the concentration of the reactants increase.
- The concept of “concentration” applies to solution and gas (pressure)



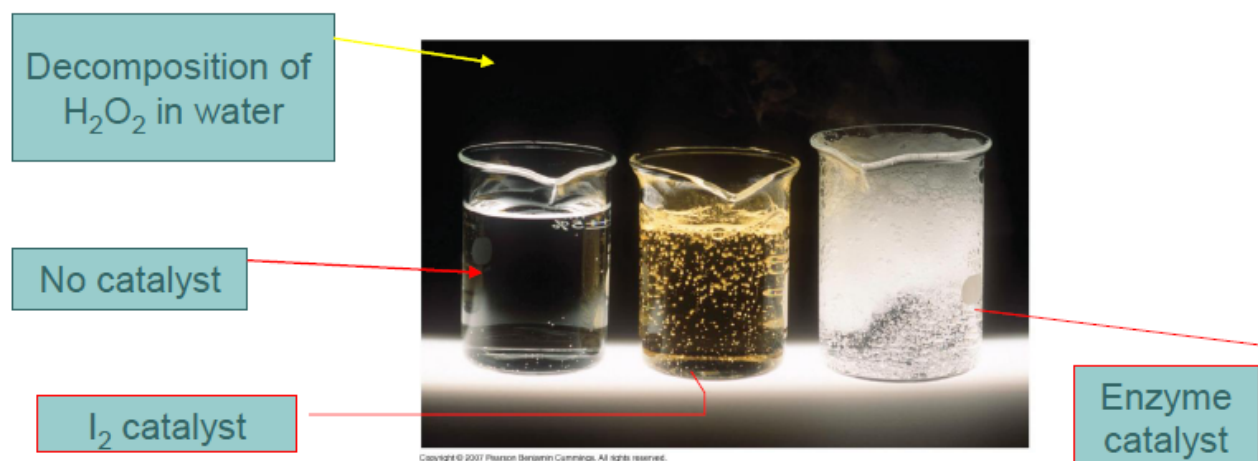
Temperature effects

- Increasing the temperature of a system increases the average kinetic energy of its constituent particles.
- The faster the particles move, the more frequently they collide, the faster the rate of reaction.
- Hence, rate of all reactions increase with increasing temperature



Catalyst effects

- A catalyst is a substance that participates in a chemical reaction and increases the rate of the reaction without undergoing a *net* chemical change itself.
- Catalysts are highly selective and often determine the products of a reaction by accelerating only one of several possible reactions that could occur.



Part 2

Rate Law

The rate law: how does concentration affects rate

- The Rate Law (or rate equation) expresses the rate as a function of reactant concentrations.
- The rate of a reaction (in simple cases) is directly proportional to the concentration of each reactant raised to a power.
- For the reaction $aA + bB \rightarrow cC + dD$ the rate law would be:

$$\text{Rate} = k[A]^m[B]^n$$

- m, n are called the *reaction order* of each reactant and are determined *experimentally* (i.e. generally no relationship with the stoichiometric coefficients of a reaction and cannot be predicted from a balanced chemical equation)
- they determine how rate is affected by the concentration of each reactant
- k is called the *rate constant*, and its value is fixed at a given temperature

Order of reaction

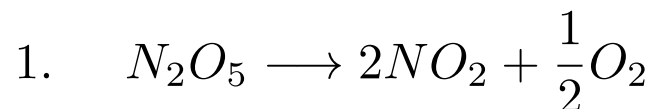
- The order of a reaction with respect to a given reactant is the power of the reactant's concentration in the experimentally determined rate equation.
- The *overall order* of a reaction is the sum of the powers of the concentration terms in the rate equation.

For the reaction $aA + bB \rightarrow cC + dD$ with rate equation:

$$\text{Rate} = k[A]^m[B]^n$$

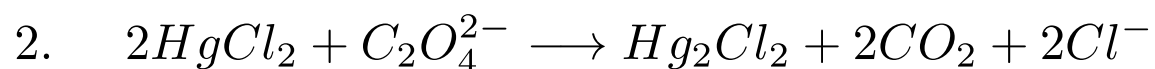
- The order of reaction with respect to reactant A is m , and with respect to reactant B is n . The overall order of the reaction is $m+n$.
- The order of a reaction is usually a positive integer, but can be negative, zero, or even a fraction.

Examples of reaction order



$$Rate = k[N_2O_5]^1$$

$$Overall\ Order = 1$$



$$Rate = k[HgCl_2][C_2O_4^{2-}]^2$$

$$Overall\ Order = 3$$



$$Rate = k[IO_3^-][H^+]^2[I^-]^2$$

$$Overall\ Order = 5$$

1st order with respect to $[IO_3^-]$

2nd order with respect to $[H^+]$ or $[I^-]$

Overall order of reaction = 1+2+2

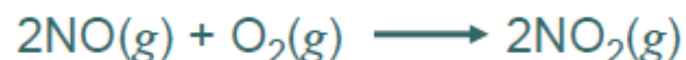
= 5th order

Zero-, first- and second-order reactions

$$\text{Rate} = k[A]^m[B]^n$$

- **Zeroth-order**: changing the concentration has no effect on the rate
 - When concentration *2, rate _____
- **First-order**: the rate changes by the same factor as the concentration
 - When concentration *2, rate _____
- **Second-order**: the rate changes by the square of the factor the concentration changes
 - When concentration *2, rate _____

Determining reaction orders – initial rate method



- Monitor the effect on the initial rate of the reaction when the initial concentration of that reactant is changed

Why initial rate?

Why not rate after sometime, such as 20 min?

Experiment	Initial Reactant Concentrations (mol/L)		Initial Rate (mol/L*s)
	O ₂	NO	
1	1.10x10 ⁻²	1.30x10 ⁻²	3.21x10 ⁻³
2	2.20x10 ⁻²	1.30x10 ⁻²	6.40x10 ⁻³
3	1.10x10 ⁻²	2.60x10 ⁻²	12.8x10 ⁻³
4	3.30x10 ⁻²	1.30x10 ⁻²	9.60x10 ⁻³
5	1.10x10 ⁻²	3.90x10 ⁻²	28.8x10 ⁻³

What is the reaction order?

Other than initial rate method?

- Initial rate method is useful, but not always applicable.
 - What is the potential problem of initial rate method?
- Integrated rate law: relationship between concentration and time
 - What is the benefit of the integrated rate law, when conducting experiments?

$$\ln[A] = -k_1 t + \ln[A]_0$$

- Differential rate laws: relationship between rate and concentration

$$\text{Rate} = -\frac{d[A]}{dt} = k_1[A]$$

Integrated rate law of reactions

Zero – order reaction :

$$rate = -\frac{\Delta[A]}{\Delta t} = k_0[A]^0$$

$$[A]_t = -k_0t + [A]_0$$

$$t_{\frac{1}{2}} = \frac{[A]_0}{2k_0}$$

First – order reaction :

$$rate = -\frac{\Delta[A]}{\Delta t} = k_1[A]$$

$$\ln [A]_t = -k_1t + \ln[A]_0$$

$$t_{\frac{1}{2}} = \frac{\ln 2}{k_1}$$

For 1st order reaction, its half life $t_{\frac{1}{2}}$ is a constant, independent of concentration.

Second – order reaction :

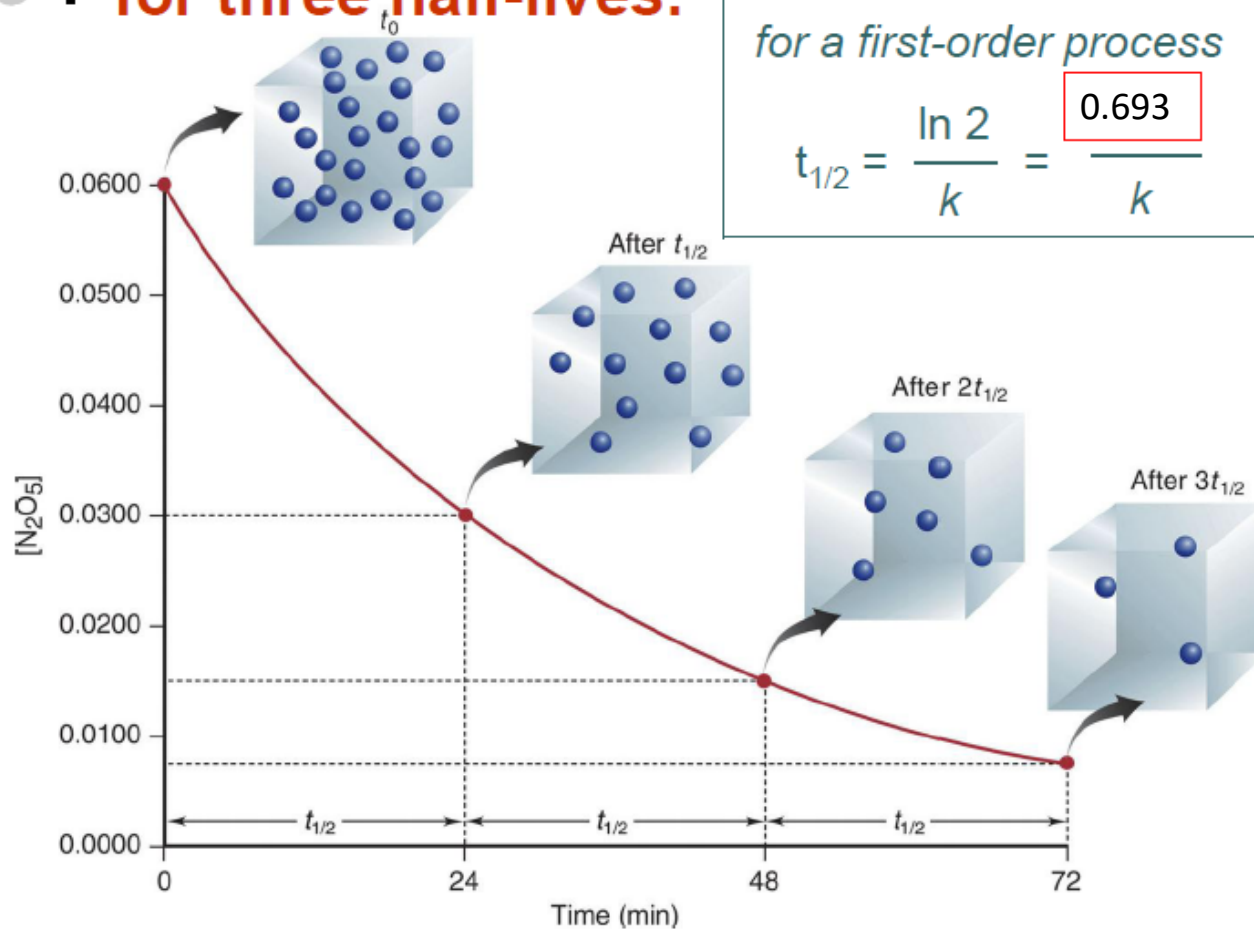
$$rate = -\frac{\Delta[A]}{\Delta t} = k_2[A]^2$$

$$\frac{1}{[A]_t} = k_2t + \frac{1}{[A]_0}$$

$$t_{\frac{1}{2}} = \frac{1}{k_2[A]_0}$$

$t_{\frac{1}{2}}$ of other order is NOT a constant, i.e. varies with concentration.

A plot of $[\text{N}_2\text{O}_5]$ vs. time for three half-lives.



for a first-order process

$$t_{1/2} = \frac{\ln 2}{k} = \frac{0.693}{k}$$

Graph of Integrated Rate Laws

Integrated rate law

Zeroth order:

$$[A] = [A]_0 - kt$$

First order:

$$\ln [A] = \ln [A]_0 - kt$$

Second order:

$$1/[A] = 1/[A]_0 + kt$$

If reaction order is unknown (to be determined), how to plot graphs to determine the reaction order, given a table of concentration and time?

	Zeroth-order	First-order	Second-order
Linear graph			
Unit of k			

Part 3

Arrhenius equation and catalysis

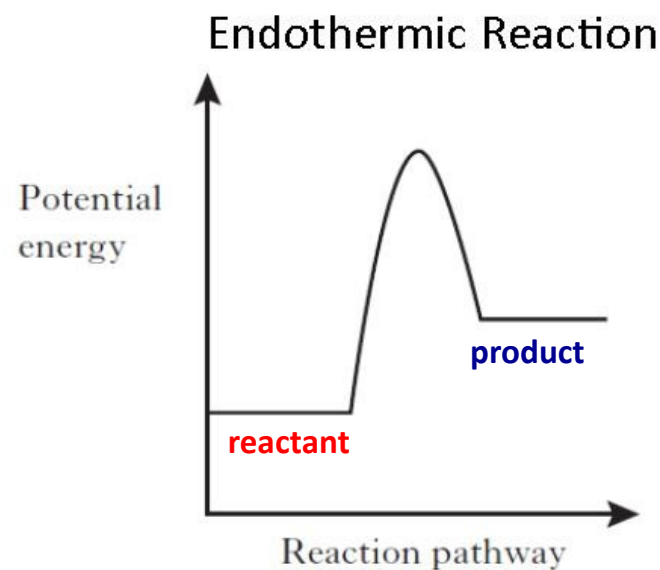
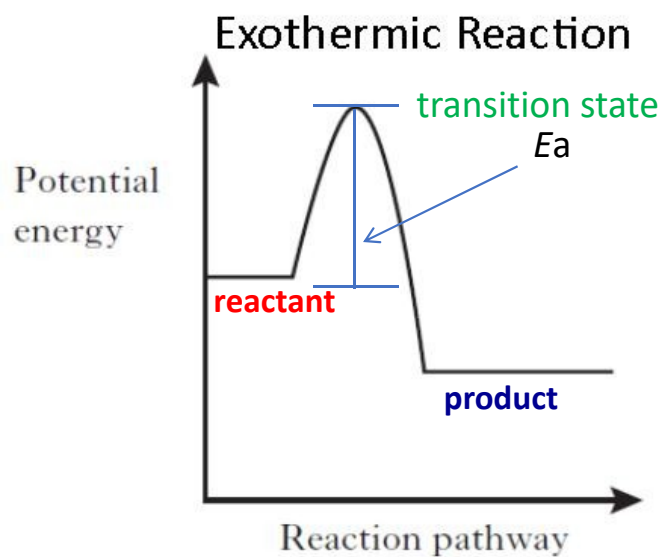
How does temperature affect rate?

- The Swedish chemist Arrhenius showed that reaction rates depend on 4 factors:
 - the **energy** of collisions between reacting molecules,
 - the **frequency of collisions**,
 - the **temperature**,
 - whether the **collisions have the correct geometry/direction**
- These are summarized by the Arrhenius equation:

$$k = Ae^{-E_a/RT}$$

E_a : activation energy

Reaction Energy Profile and Activation Energy



Arrhenius equation

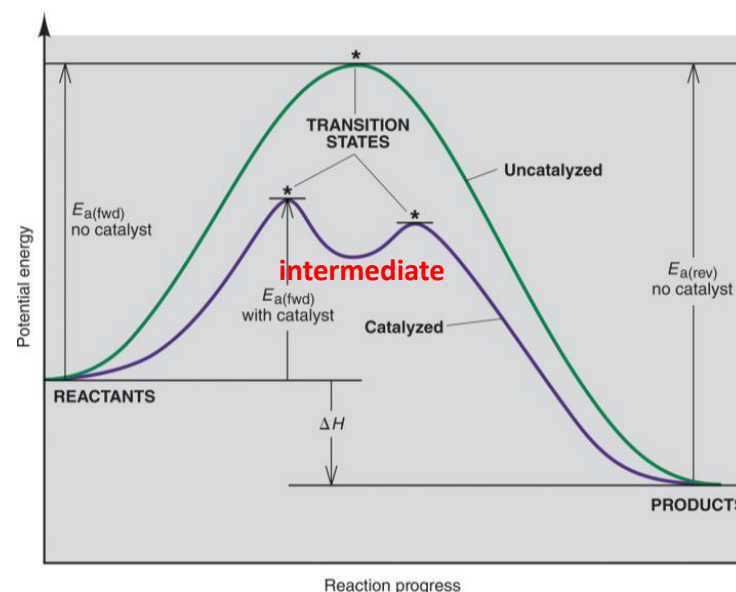
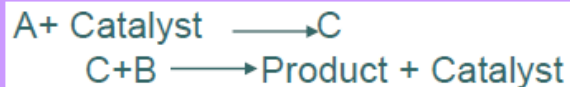
- Arrhenius equation is in an exponential form. Is it possible to change it to a linear form?

$$k = Ae^{-E_a/RT}$$

- How to calculate k_2 (at T_2), if k_1 (at T_1) is known?
- How to calculate E_a ?

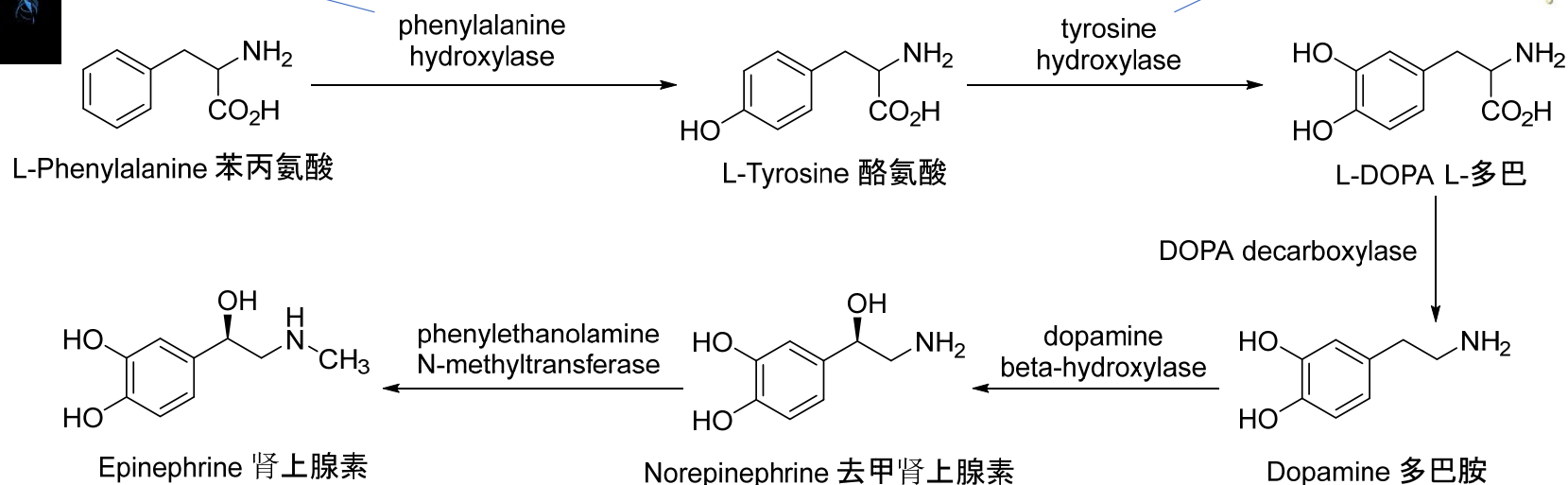
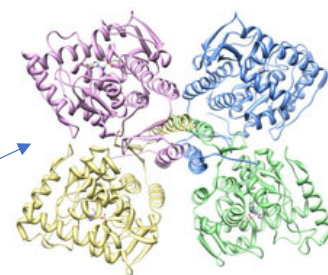
Catalysts

- A catalyzed reaction yields the products more quickly, but **DOES NOT** yield more product than the uncatalyzed reaction.
- A catalyst lowers E_a by providing a different mechanism, for the reaction through a new, lower energy pathway.
- Lowering the E_a increases the rate constant k , *and thereby increases the rate of the reaction.*
- A catalyst increases the rate of the *forward AND the reverse reactions.*



Catalysts in Nature

- Enzymes are incredibly efficient catalysts (catalyze 10^6 reactions / s)
 - They increase the reaction rate by lowering the activation energy
 - They are usually very specific



Catalysts in Nature

- There are two main models of enzyme action
 - *Lock and key model*
 - *Induced fit model*

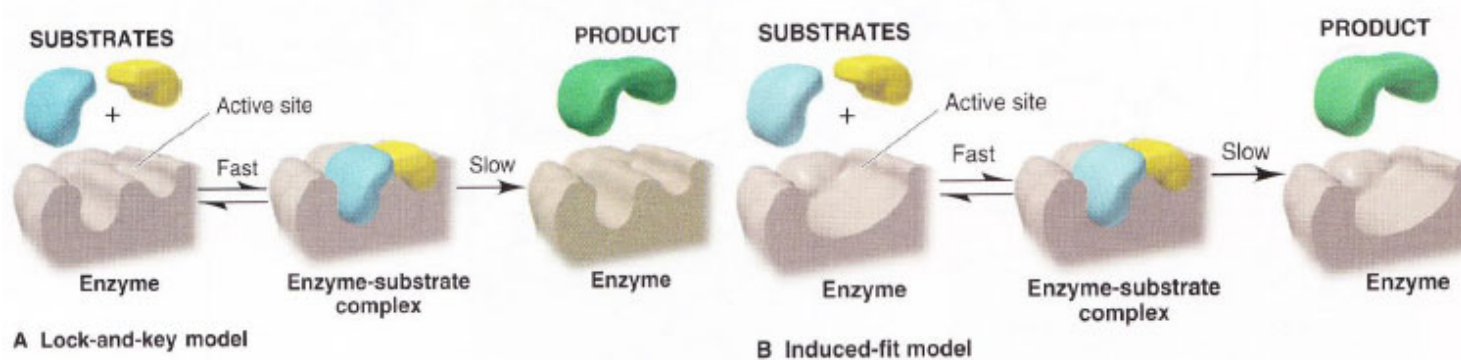
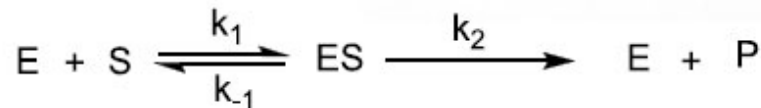


Figure B16.5 Two models of enzyme action. **A**, In the lock-and-key model, the active site is thought to be an exact fit for the substrate shapes. **B**, In the induced-fit model, the active site is thought to change shape to fit the substrates. Most enzyme-catalyzed reac-

tions proceed through the fast, reversible formation of an enzyme-substrate complex, followed by a slow conversion to product and free enzyme.



Michaelis-Menten equation

$$v = \frac{V_{max}[S]}{K_m + [S]}$$